A MORPHISM OF ALGEBRAIC SPACES CAN BE LIFTED TO CHOSEN PRESENTATIONS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file spaces.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The statement contains an xymatrix display; it is part of the counted statement span and is preserved verbatim, with the chapter's own Xy-pic setup retained in the preamble.

Editorial note. The two references of the counted spans are recorded as non-load-bearing navigation pointers; the argument that consumes them stays inside the counted proof span.

Editorial note. Preamble: sources/stacks-spaces/retained-preamble.tex, sliced mechanically from the chapter's own preamble.tex.

Allowed standard prerequisites (editorial).

- Base change keeps surjectivity and etaleness. Lemma lemma-base-change-representable-transformations-property, used at proof lines 2088-2090. Makes U -> W surjective and etale.
- Composition of representable transformations with a property. Lemma lemma-composition-representable-transformations-property, used at proof lines 2090-2093. Makes U -> F surjective and etale.
- Definition of an algebraic space. used at proof lines 2086-2087. Identifies the fibre product W x_G V with a scheme.

Source reference. Original lemma 02WL: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 3808–3808. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 02WK: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 3807–3807. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Lemma 0.2. *Let S be a scheme contained in Sch_{fppf}. Let F, G be algebraic spaces over S. Let $a : F \to G$ be a morphism. Given any $V \in \mathrm{Ob}((Sch/S)_{fppf})$ and a surjective étale morphism $q : V \to G$ there exists a $U \in \mathrm{Ob}((Sch/S)_{fppf})$ and a commutative diagram*

$$\begin{array}{ccc} U & \xrightarrow{\alpha} & V \\ {\scriptstyle p}\downarrow & & \downarrow{\scriptstyle q} \\ F & \xrightarrow{a} & G \end{array}$$

with p surjective and étale.

Date: stacks-project:spaces.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Proof. First choose $W \in \mathrm{Ob}((Sch/S)_{fppf})$ with surjective étale morphism $W \to F$. Next, put $U = W \times_G V$. Since G is an algebraic space we see that U is isomorphic to an object of $(Sch/S)_{fppf}$. As q is surjective étale, we see that $U \to W$ is surjective étale (see Lemma 02WL). Thus $U \to F$ is surjective étale as a composition of surjective étale morphisms (see Lemma 02WK). $\square$